Day 3 Data Report

Multi-sensor self-monitoring, 19 October 2026,
with comparison to days 1–2 and to participant 1

Participant: Spriha (participant 2)

Analysis generated with `pipeline/daily_analysis.py`,
`compare_days.py` and `compare_participants.py`

1 Summary

Day 3 contains the bedroom night of 18–19 October (00:48–09:01), an 11-minute midday outing with both insoles (a short walk, a 5-minute rickshaw ride and a short walk), almost three hours of sitting with the posture cushion, three logged drinks and the wrist band, which now records heart rate every minute. As on days 1–2, all three right-insole pressure sensors are replacements of a different type and are normalised before the feet are compared. Table 1 lists the headline metrics. The main findings are:

- **Bedroom**: a hot night, 32–34 °C (corrected) throughout, with $PM_{2.5}$ of 82–88 $\mu g/m^3$ until about 02:00 and 63–72 afterwards, the same early-night shape as on all of participant 1's nights. The band recorded only 2 h of sleep (03:01–05:03).
- **Third rickshaw ride** (12:09:35–12:14:50, recognised from foot unloading and vibration): $PM_{2.5}$ 76 $\mu g/m^3$ (peak 147), 75 dBA. Unlike day 2, heart rate fell during the ride (88–94 bpm) after 114–121 bpm just before setting off.
- **Left versus right** (Section 5): stride time and the timing of the foot's rotation were the same on both feet (tilt peak 65 vs 63–64 % of the stride). The left foot again rotated further (tilt range 55–64 vs 46–48°), now on all three days. Pressure stance and IMU timing differences again follow the replaced sensors and the half-rate right insole.
- **Drinking**: 1,075 mL logged: 500 mL at 00:06, then nothing for 19 hours until 500 mL at 19:09 and 75 mL at 20:02.
- **Sitting posture**: 95 of 171 minutes monitored; 55 % "straight", 24 % "lean forward" and 18 % leaning back, a more relaxed pattern than on day 2 (99 % straight or forward).

Table 1: Headline metrics of day 3 (19 October 2026).

Domain	Metric	Value	Unit
Sleep	Time asleep	2.0	h
	Sleep efficiency	100	%
	Light / deep / REM	100 / – / –	%
	Mean HR asleep	70	bpm
	Mean SpO_2 asleep	nan	%
Activity	Steps (band / phone)	3,731 / –	
	Mean awake HR	95	bpm
	Minutes with HR $>$ 120 bpm	19	min
	HR–steps Spearman ρ	0.28	
	HR lag behind steps	2	min
Gait (walk)	Cadence	108	steps/min
	Stride time / CV	1.11 / 11.8	s / %
	Stance, IMU (R / L)	65.0 / 67.7	%
	Swing, IMU (R / L)	0.405 / 0.359	s
	Stance, pressure (R)	–	%
	Push-off timing (R / L)	68.6 / 70.2	%
	Distance / speed	41 / nan	m / m/s
	Stride length	nan	m
	Insole − band bias	+27.5	steps/min
Posture	Monitored time	95	min
	Upright / slouched	79 / –	%
	Median time between changes	1.8	min
Environment	Indoor CO_2 (median)	–	ppm
	Indoor / outdoor $PM_{2.5}$	– / 79	$\mu g/m^3$
	Indoor temp. / RH	–	°C / %
	Indoor / outdoor sound	– / 69	dBA
	Bedroom CO_2 / $PM_{2.5}$	340 / 69	ppm / $\mu g/m^3$
	Bedroom temp. / sound	32.6 / 49	°C / dBA
Left/right	Left lag behind right	22	% of cycle
	Foot-tilt range L − R	-3.0	°
	Stance / swing SI (IMU)	–	%
	Stance % (L / R, IMU)	–	%
Hydration	Total intake (drinks)	1075 / 3	mL
	Longest gap between drinks	19.0	h

2 Data collected

Table 2: Day 3 recordings (local time, UTC+6).

Key	Session	Time	Sensors	Notes
O	Bedroom	00:48–09:01	Environment	Night of 18–19 October
W	Walk	12:07–12:09	Insoles, environment, GPS	65 strides per foot
K	Rickshaw ride	12:09–12:15	Environment	Seated, vibrating
A	Walk after the ride	12:15–12:18	Insoles, environment, GPS	77–89 strides
P	Sitting	14:26–17:17	Posture cushion	Cushion connected 95 min; no environment node

The upload contained two folders for the midday outing, `Day-3 mor` and `Day_3 afternoon`; they are byte-for-byte identical (12:07–12:18), so the recording is analysed once and no separate afternoon outing is available. The night recording `Night_3` is the same file that was earlier filed with day 2; it is analysed here, and the day-2 report no longer includes it (a nested duplicate folder is ignored). The ride was not noted; from 12:09:35 both insoles' load fell (the right to about half), vibration continued until 12:13:50, she sat still until 12:14:40 and walked off at 12:14:50. The right insole streamed at about 62 samples/s, the left at 114. The band now records heart rate every minute (507 of 565 readings at 1-minute intervals). Drinks from the bottle log: 500 mL at 00:06, 500 mL at 19:09 and 75 mL at 20:02. Temperatures include the $-5\,^{\circ}\mathrm{C}$ correction.

3 Bedroom, night of 18–19 October

The bedroom (Fig. 1) was hot all night: 32–34 °C after the correction (hourly medians), with 46–50 % humidity. $PM_{2.5}$ was 82–88 $\mu g/m^3$ until about 02:00 and 63–72 afterwards, the same early-night pattern seen on all of participant 1's nights, though at a lower level. The sound level fell from about 59 dBA before 02:00 to 47–49 dBA. CO_2 stayed at 314–371 ppm; since the sensor reads low (below the outdoor level of about 420 ppm), only changes are meaningful, and there was no build-up overnight. The band recorded sleep only from 03:01 to 05:03 (2.0 h, all light sleep, heart rate 70 bpm; Fig. 2), so it probably missed most of the night; the correlations between heart rate and the bedroom (Fig. 3) rest on those two hours only.

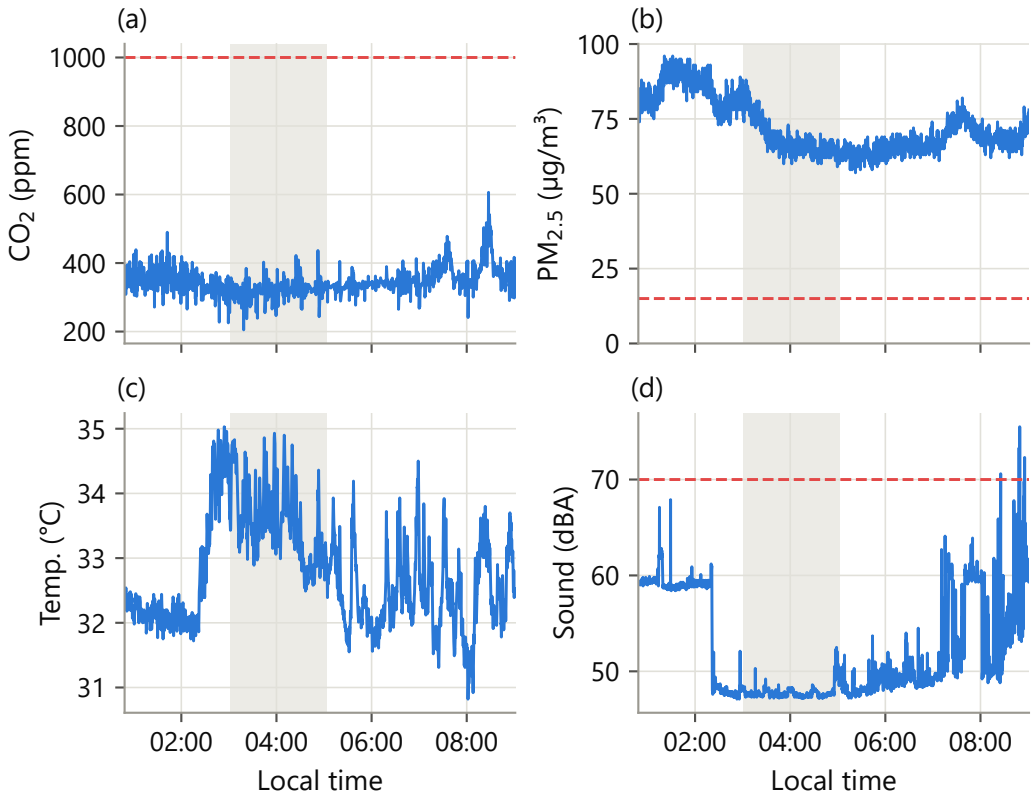


Figure 1: Bedroom environment overnight (session O, 00:48–09:01, 10 s averages): (a) CO_2, median 340 ppm (max 606); (b) $PM_{2.5}$, median 69 μg/m^3 (max 96); (c) temperature, median 32.6 °C; (d) sound level, median 49 dBA. Dashed lines: 1000 ppm CO_2, WHO $PM_{2.5}$ 24-h guideline, 70 dBA. The grey band marks sleep (03:01–05:03) from the wrist band. Temperature includes a -5 °C correction for sensor self-heating.

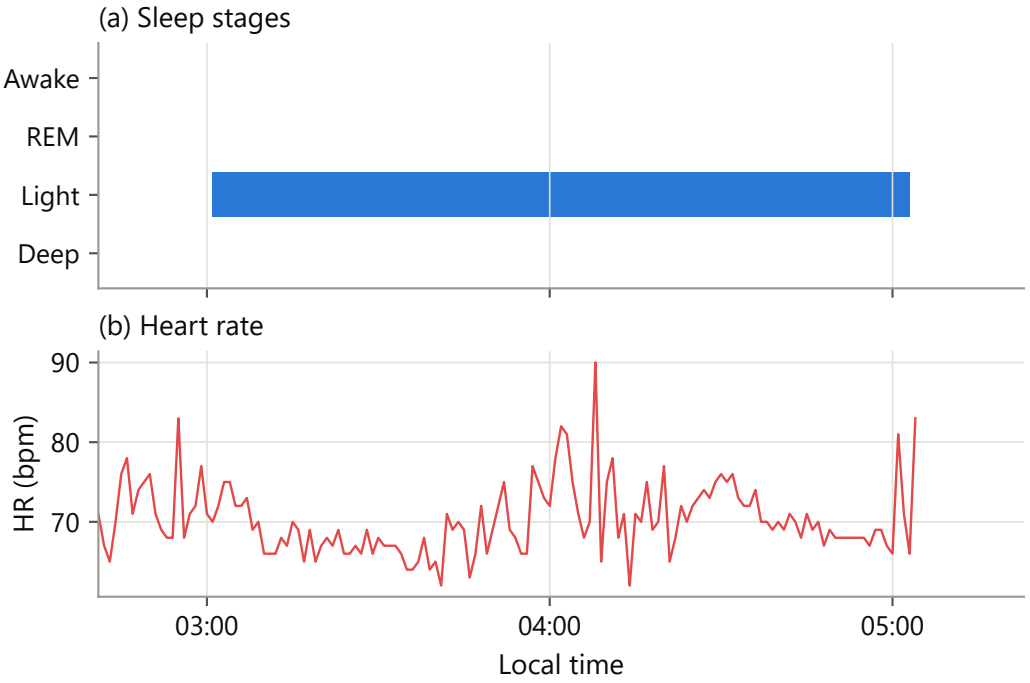


Figure 2: Sleep before day 3 from the wrist band. (a) Hypnogram, 03:01–05:03: 2.0 h asleep (efficiency 100 %), 100 % light, 0 % deep and 0 % REM. (b) Heart rate (mean 70 bpm, lowest 10-min mean 65 bpm). The band recorded no SpO_2.

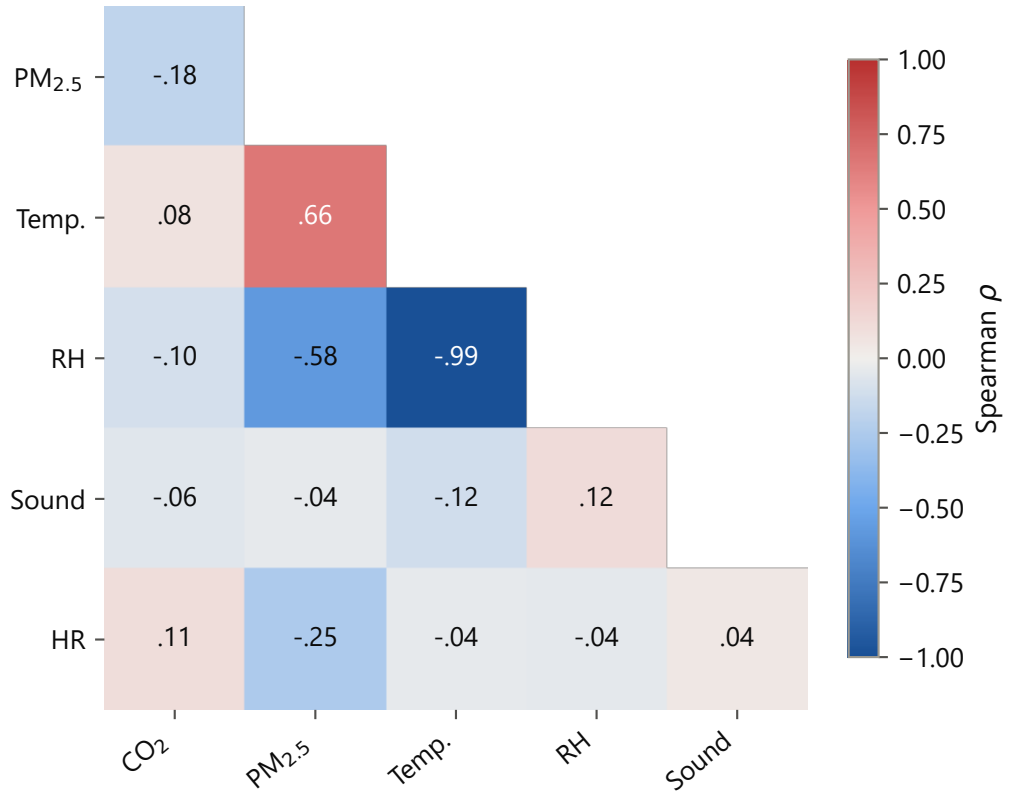


Figure 3: Correlations between bedroom environment and heart rate during sleep (session(s) O, sleep minutes only). Spearman rank correlations of 1-min means ($n = 123$–123 minutes per pair). Minute samples are autocorrelated, so coefficients are descriptive and no p-values are reported.

4 Midday outing and rickshaw ride

She walked for about two minutes (12:07–12:09; 65 strides per foot, stride time 1.06 s, cadence 114 steps/min), rode the rickshaw for five minutes and walked again for about three and a half minutes (12:15–12:18; stride 1.11–1.13 s, cadence 107–108 steps/min; Figs. 4–8). Stance from the left pressure sensors was 62 % of the stride in both walks. GPS had only a few fixes, so distance and speed are not available. In the six minutes with insole data the insoles counted more steps than the band (300 vs 135), as the band undercounts short walks (Fig. 9).

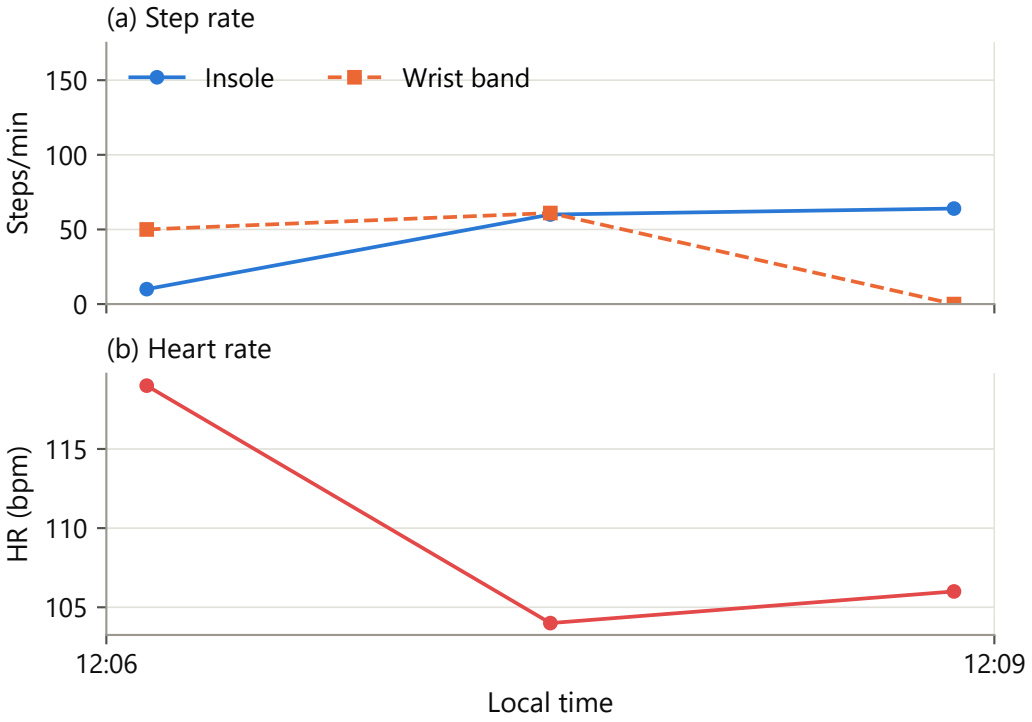


Figure 4: Session W (walk before the ride) minute by minute: (a) step rate from the insoles (2 × the strides of whichever insole detected more in that minute) and the wrist band; (b) wrist-band heart rate (mean 110 bpm).

The ride (Fig. 10) had $PM_{2.5}$ of 76 $\mu g/m^3$ (median; peak 147), 75 dBA and 32 °C, against 70–79 $\mu g/m^3$ on the walks. Heart rate, now recorded every minute, was 114–121 bpm just before she set off (12:04–12:07), 104–107 bpm while walking and fell to 88–94 bpm during the ride, then rose again to 118 bpm in the walk after it. On day 2 her heart rate had been highest in the ride (118–122 bpm); here the seated ride let it come down. The difference

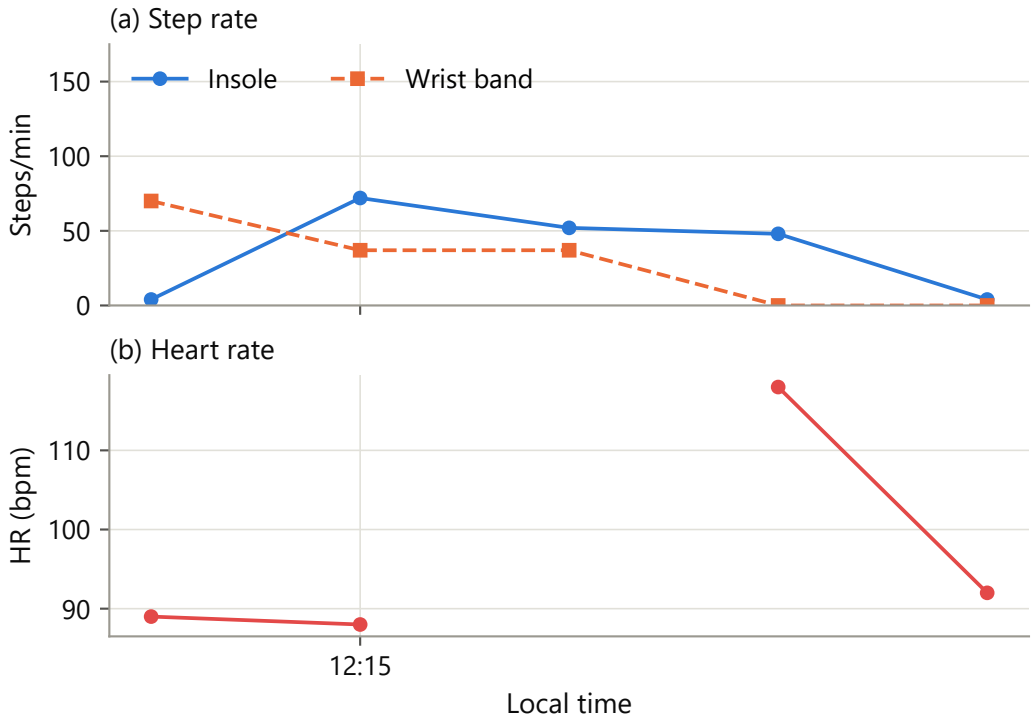


Figure 5: Session A (walk after the ride) minute by minute: (a) step rate from the insoles (2 × the strides of whichever insole detected more in that minute) and the wrist band; (b) wrist-band heart rate (mean 97 bpm).

between the two days probably reflects what came before each ride (day 2: 11 minutes standing in the heat; day 3: a burst of activity just before leaving).

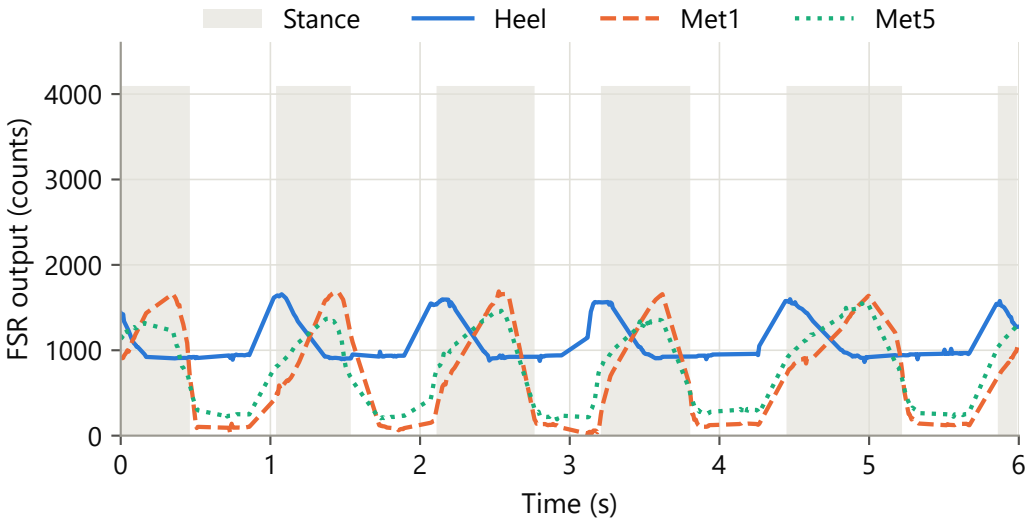


Figure 6: Right-insole plantar pressure during 6 s of steady walking (session A, walk after the ride). Lines show the heel, first-metatarsal (Met1) and fifth-metatarsal (Met5) force-sensitive resistors (12-bit ADC counts; higher means more load). Shaded intervals are stance phases detected by hysteresis thresholding of the summed, 6 Hz low-pass filtered load.

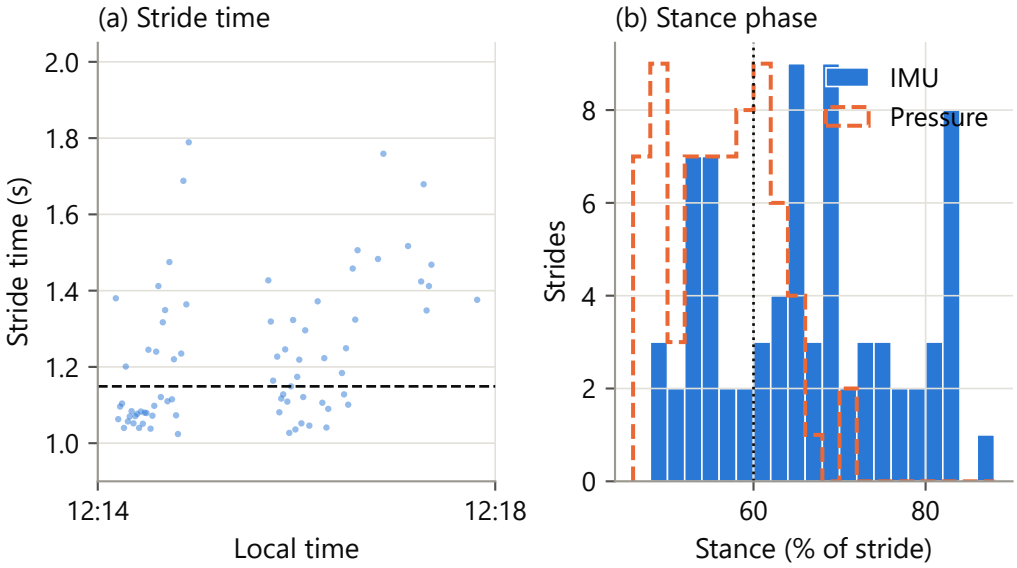


Figure 7: Right-foot stride parameters, session A (77 strides). (a) Stride time of every detected stride; dashed line = median (1.15 s, cadence 108 steps/min). (b) Stance phase as a percentage of stride time from IMU gait events (bars; heel-strike impact to IMU toe-off; median of steady-walking strides 65.0 %) and from the pressure sensors (dashed outline; 54.8 %) – not valid on this day because the first-metatarsal sensor had been replaced; dotted line = typical healthy value of about 60 %.

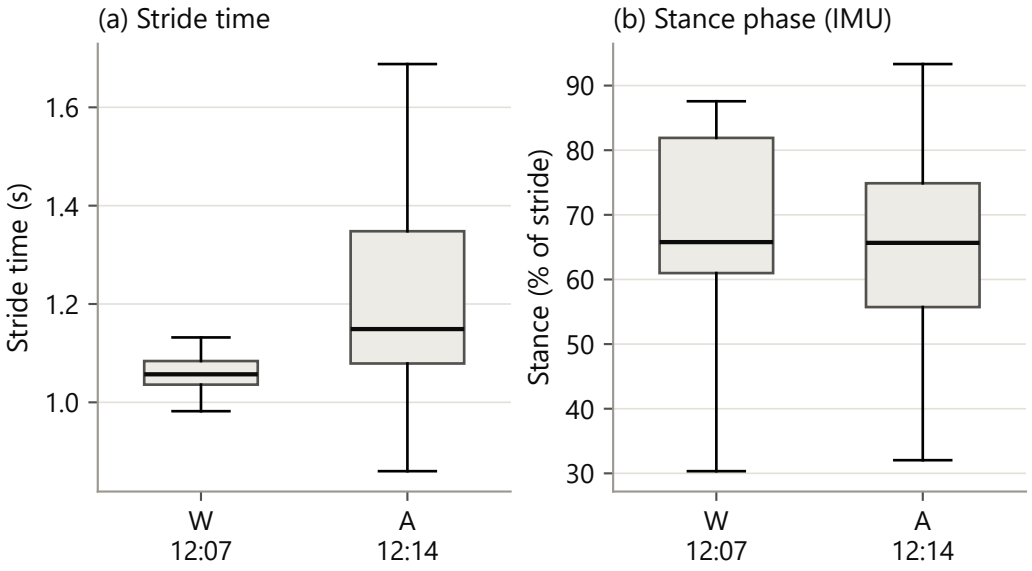


Figure 8: Right-foot (a) stride time and (b) stance phase (IMU gait events) across the gait recordings of day 3 (W: $n = 65$, A: $n = 77$; session start time below each key). Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR (outliers not drawn).

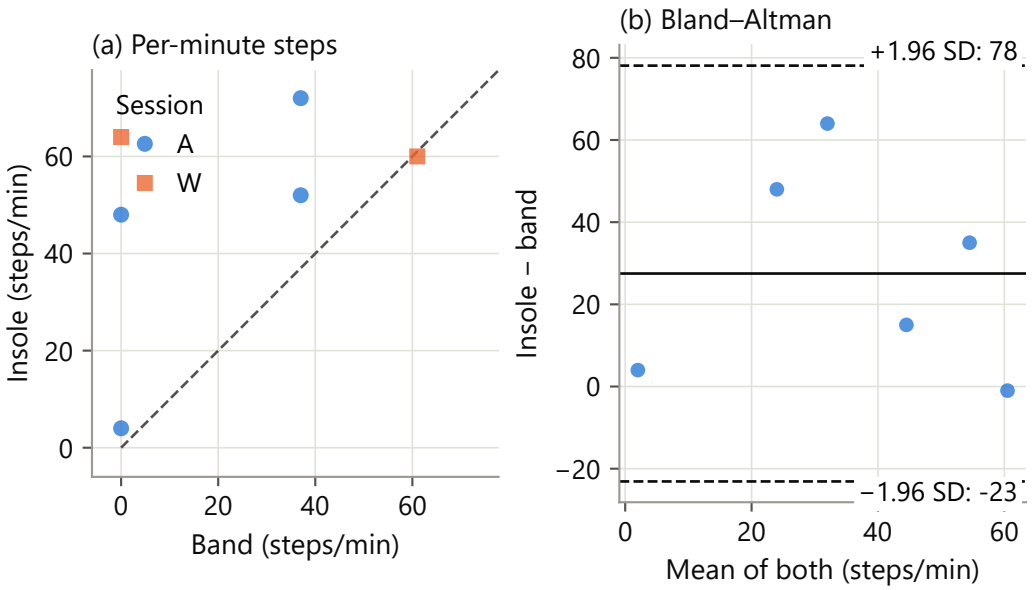


Figure 9: Agreement between insole-derived steps ($2 \times$ the strides of the insole with more detections in each minute) and wrist-band steps for every minute of the gait sessions (6 minutes; totals 300 vs 135). (a) Per-minute counts with the line of identity ($r = 0.48$). (b) Bland–Altman plot: mean difference +27.5 steps/min (solid), 95 % limits of agreement -23 to 78 steps/min (dashed).

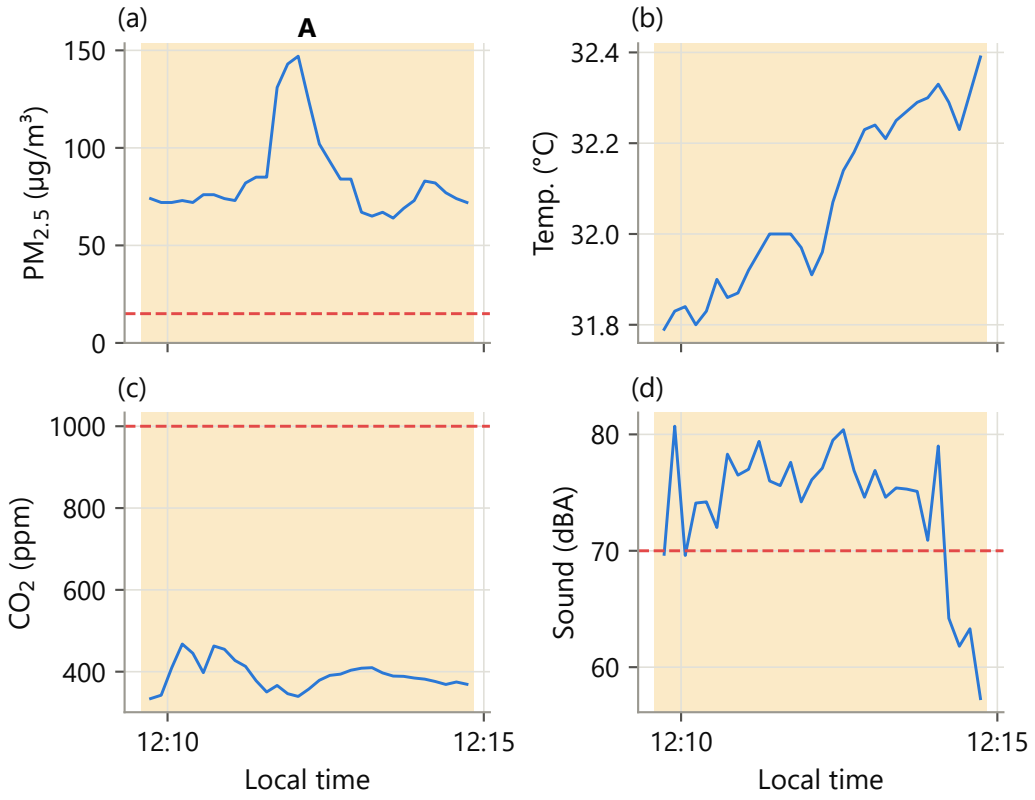


Figure 10: Environment node during session K (rickshaw ride, 10 s averages): (a) $PM_{2.5}$, median 76 μg/m^3, peak 147 (dashed: WHO 24-h guideline); (b) temperature, median 32.0 °C; (c) CO_2; (d) sound level, median 75 dBA (dashed: 70 dBA). Panel (c) shows CO_2 instead of humidity (median 389, maximum 468 ppm); the sensor reads low outdoors, so the rise inside vehicles is more reliable than the absolute level. Yellow bands mark noted events: A rickshaw ride (from foot load and vibration) (12:09–12:14). Temperature includes a -5 °C correction for sensor self-heating.

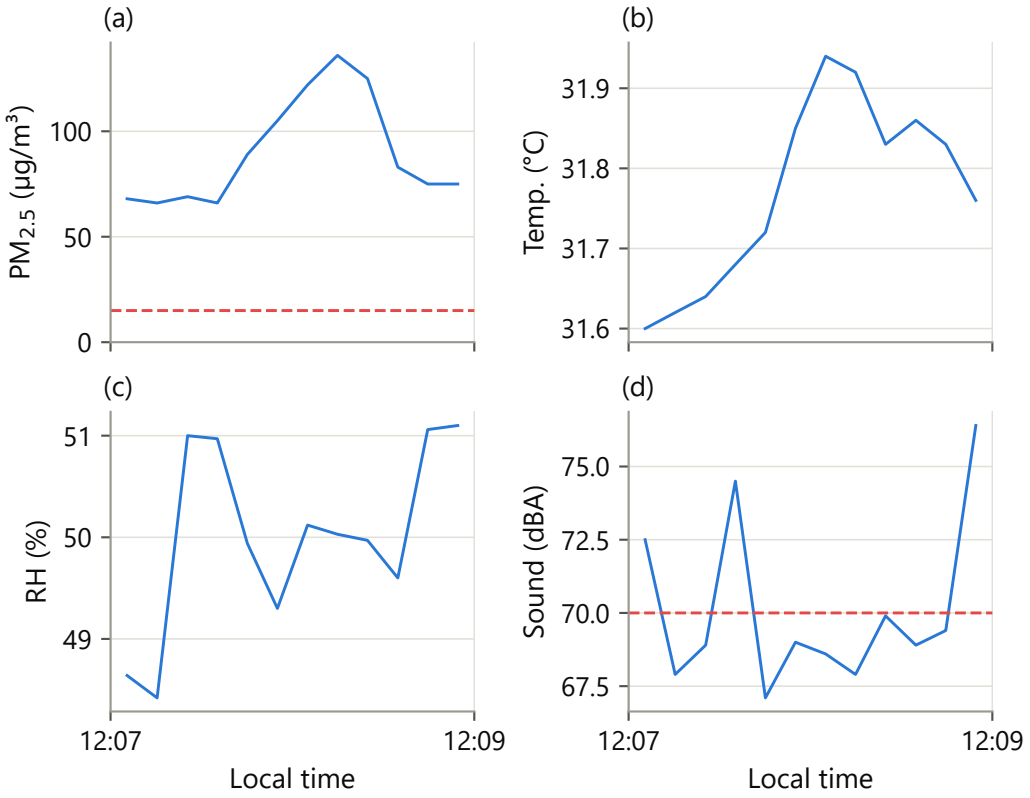


Figure 11: Environment node during session W (walk before the ride, 10 s averages): (a) $PM_{2.5}$, median 79 μg/m^3, peak 136 (dashed: WHO 24-h guideline); (b) temperature, median 31.8 °C; (c) relative humidity, median 50 %; (d) sound level, median 69 dBA (dashed: 70 dBA). CO_2 is omitted because readings fell below the physically plausible outdoor level of about 420 ppm (minimum 340 ppm). Temperature includes a -5 °C correction for sensor self-heating.

5 Left versus right foot

The right insole again streamed at half rate, so the insoles' clocks are not aligned and fewer than 100 strides could be paired; the feet are compared from each foot's own strides (Figs. 12 and 13; Table 3).

Table 3: Left and right foot in the two walks of day 3, from each foot's own steady strides. [a]The replacement right sensors respond at a different force. [b]Follows which insole streamed at full rate (left on day 3).

	Walk W		Walk A		
Measure	Left	Right	Left	Right	Valid
Stride time (s)	1.06	1.06	1.12	1.10	yes
Foot-tilt peak (% of stride)	65	64	65	63	yes
Foot-tilt range (°)	64	48	55	46	yes
Load share heel/Met1/Met5 (%)	23/38/39	38/28/34	25/36/39	40/28/31	no[a]
Pressure stance (%)	61.6	55.2	62.3	55.0	no[a]
IMU push-off (%)	72.6	67.9	70.2	68.5	no[b]
IMU stance (%)	70.9	66.9	67.7	65.0	no[b]

- *The same on both feet*: stride time and the point of the stride at which the foot is most rotated (tilt peak at 63–65 % on both feet, as on day 1).
- *The left foot rotates further*: tilt range 55–64° on the left against 46–48° on the right, a larger difference than on days 1 and 2 (left 4–8 % larger). Seen on all three days, this is the most consistent left-right difference in her data; it may reflect her push-off or how each insole's IMU sits in the shoe.
- *Load share on the right is not usable today*: the replacement right heel sensor never fell below 30 % of its range during swing (Fig. 12a), so it appears loaded all stride and inflates the right heel share (38–40 % vs 23–25 % on the left). On days 1–2 the shares were similar on both feet.
- *Pressure stance and IMU timing* differ for the same reasons as on days 1–2: the right sensors release earlier (stance 55 vs 62 %) and the IMU push-off is larger on the full-rate (left) insole.

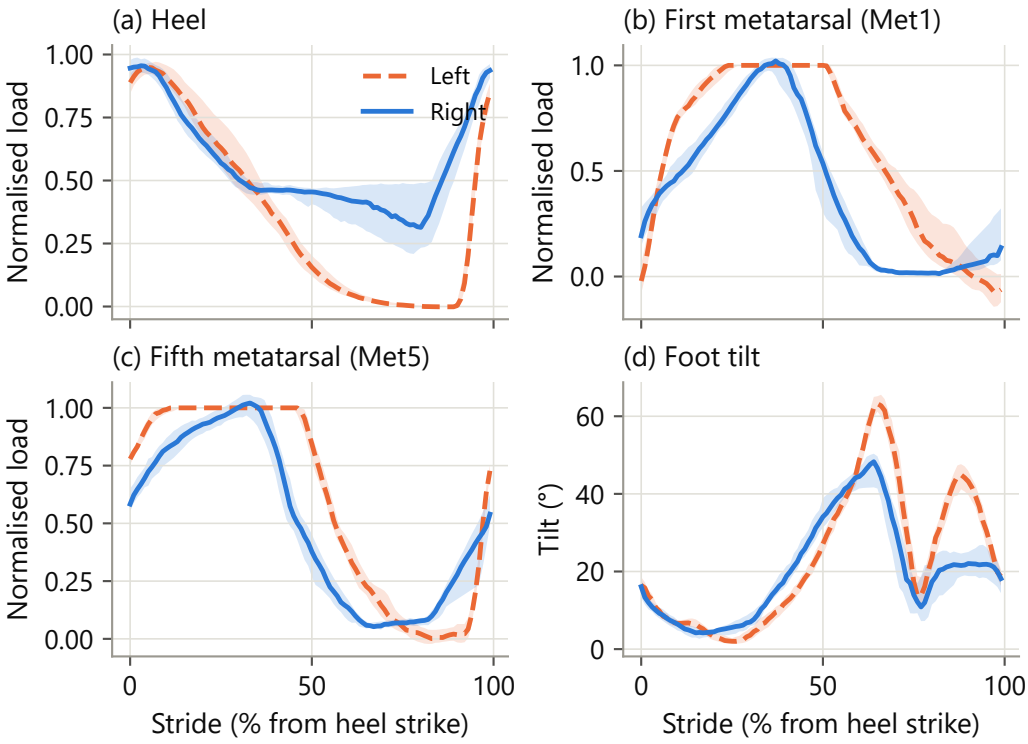


Figure 12: Left and right foot during session W from each foot's own steady strides (left 55, right 52). Each pressure sensor is scaled between its own unloaded level (5th percentile, 0) and its 95th percentile (1), so sensors of different types and constant preload do not affect the comparison of shape and timing; lines: median, bands: inter-quartile range. (a)–(c) Heel, first- and fifth-metatarsal load over the stride; (d) foot tilt from the flat, loaded position. Pressure-based stance (normalised): left 61.6 %, right 55.2 % of the stride; stride time 1.06 vs 1.05 s; foot-tilt range 64° vs 48°.

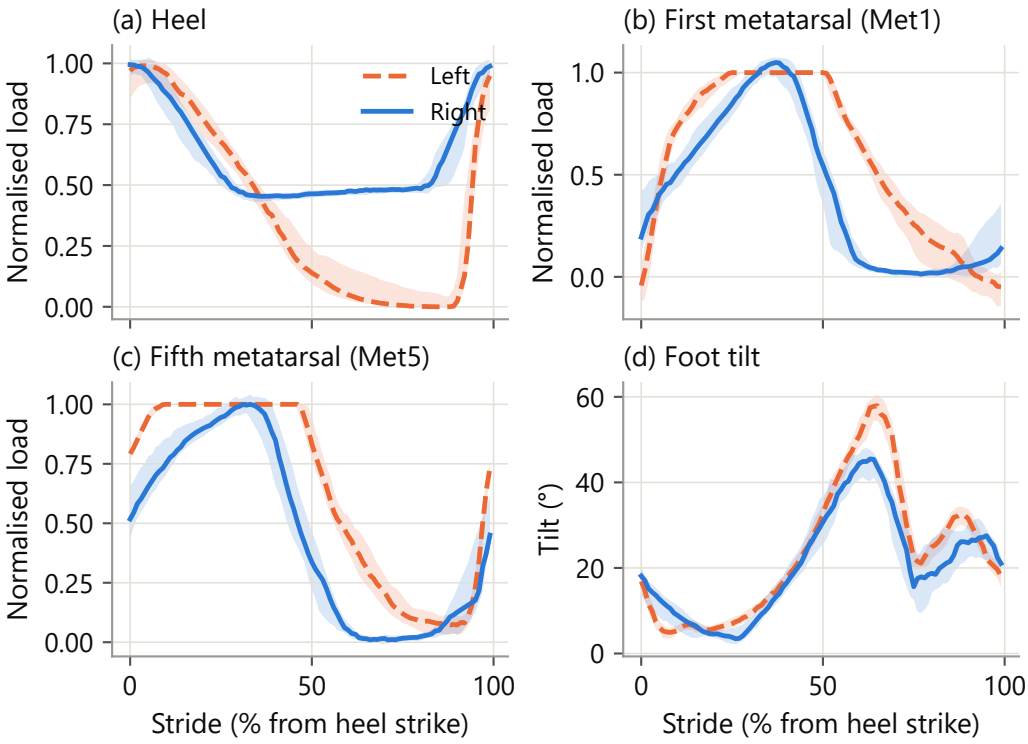


Figure 13: Left and right foot during session A from each foot's own steady strides (left 59, right 50). Each pressure sensor is scaled between its own unloaded level (5th percentile, 0) and its 95th percentile (1), so sensors of different types and constant preload do not affect the comparison of shape and timing; lines: median, bands: inter-quartile range. (a)–(c) Heel, first- and fifth-metatarsal load over the stride; (d) foot tilt from the flat, loaded position. Pressure-based stance (normalised): left 62.3 %, right 55.0 % of the stride; stride time 1.11 vs 1.10 s; foot-tilt range 55° vs 46°.

6 Sitting posture and heart rate

The cushion was connected for 95 of the 171 minutes from 14:26 to 17:17 (Fig. 14). Of the classified time, 55 % was “straight”, 24 % in the cushion’s lean-forward category, 18 % leaning back and 3 % leaning left, with a median of 1.8 minutes between changes; on day 2 she had sat almost only straight or forward (99 %). No environment data were recorded during this session.

Awake heart rate averaged 95 bpm (65–134 bpm; 19 minutes above 120 bpm; Fig. 15). By context (Fig. 16) the median was 69 bpm asleep, 92 bpm sitting indoors, 93 bpm in other awake time and 104 bpm walking. During the sitting session heart rate rose from about 85 bpm (15:00–16:00) to 102–112 bpm (16:00–17:20). With 1-minute data the heart rate followed her step rate clearly: median 89 bpm in minutes without steps, rising to 113 bpm in minutes with more than 90 steps ($\rho = 0.28$, 393 awake minutes; Fig. 17), as for participant 1.

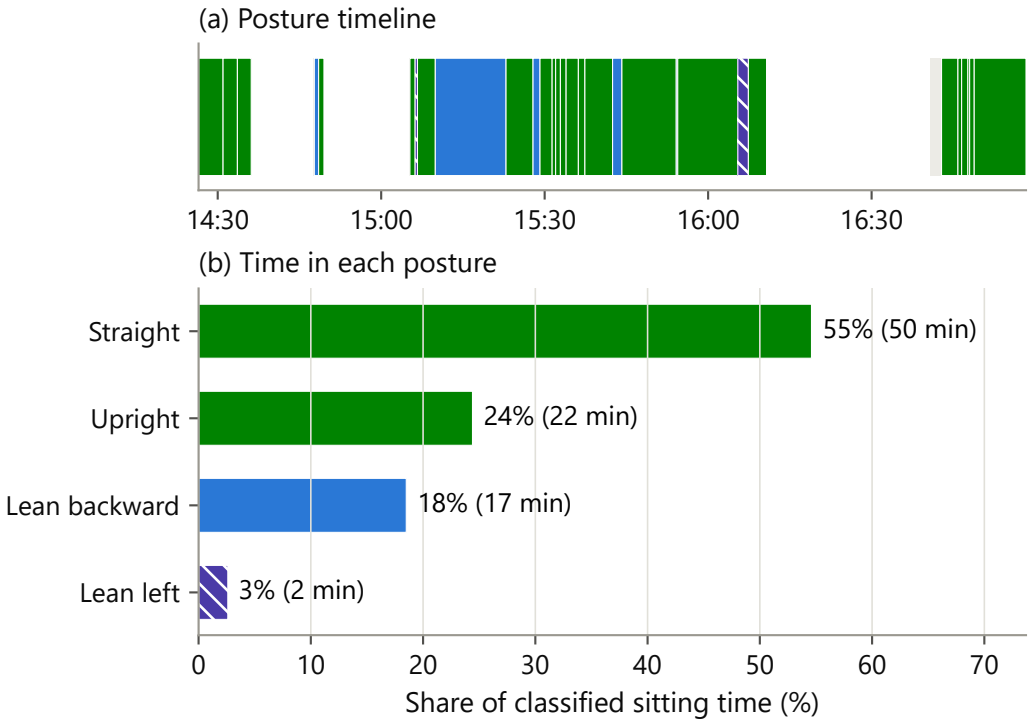


Figure 14: Sitting posture from the smart cushion. (a) Pose held over time; the cushion was connected during 14:26–14:36, 14:47–14:49, 15:05–16:10, 16:40–16:58 (95 of 171 min). Light grey = connected but no live pose event yet; white = disconnected. Onsets are shifted 10 s earlier to correct for the cushion's commit delay. (b) Share of the 92 classified minutes in each pose; median time between pose changes 1.8 min. "Upright" is the cushion's lean-forward category, which the participant confirmed is how it reports deliberately upright sitting; genuine forward leaning cannot be separated from it.

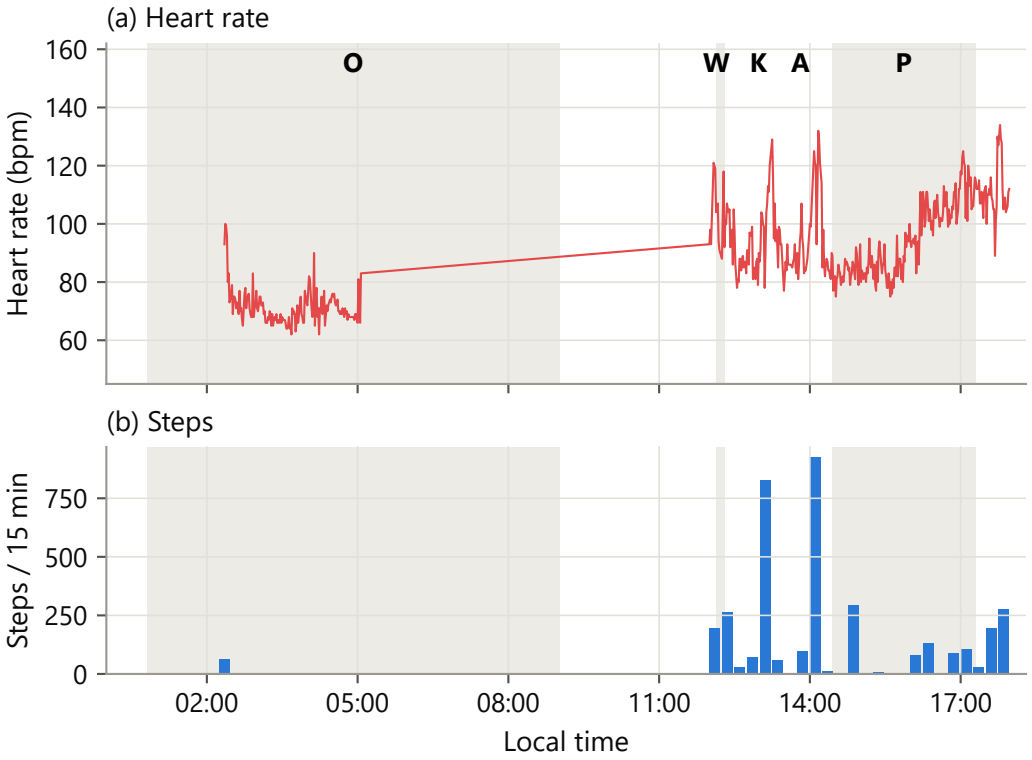


Figure 15: Wrist band over day 3: (a) heart rate (1 sample/min) and (b) steps per 15 min; sleep 03:01–05:03. Grey bands mark the sensor sessions: O bedroom, night of 18-19 october (00:48–09:01), W walk before the ride (12:07–12:09), K rickshaw ride (12:09–12:14), A walk after the ride (12:14–12:18), P afternoon sitting (posture cushion) (14:26–17:17).

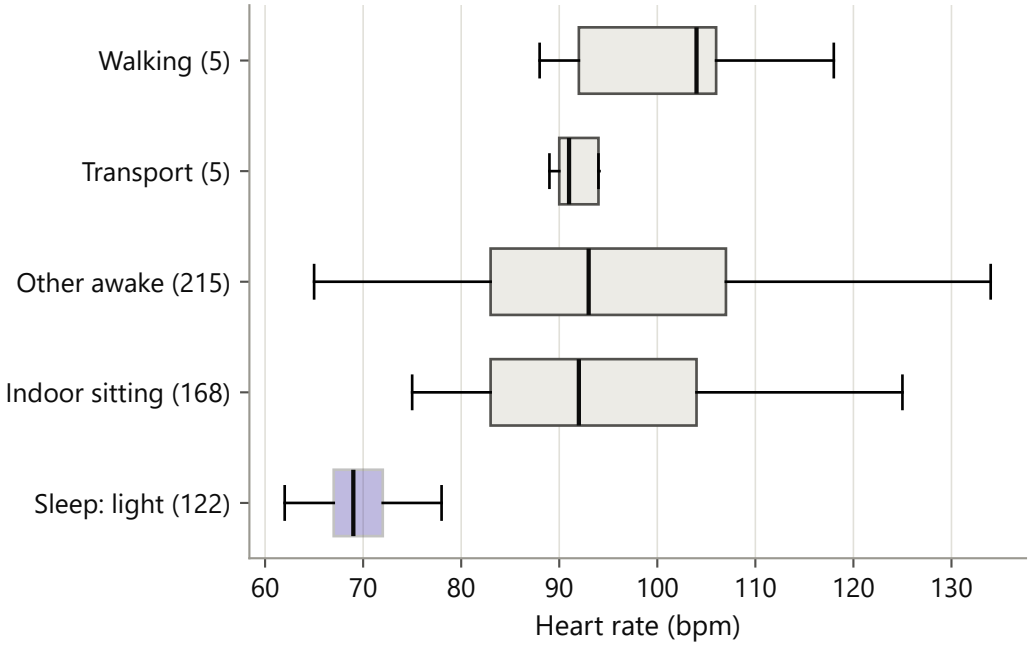


Figure 16: Heart rate by context over the day (minutes per context in brackets). Sleep stages come from the wrist band; awake contexts come from the sensor sessions (gait tests, indoor sitting with the environment node, walking sessions) and all remaining awake minutes. Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR.

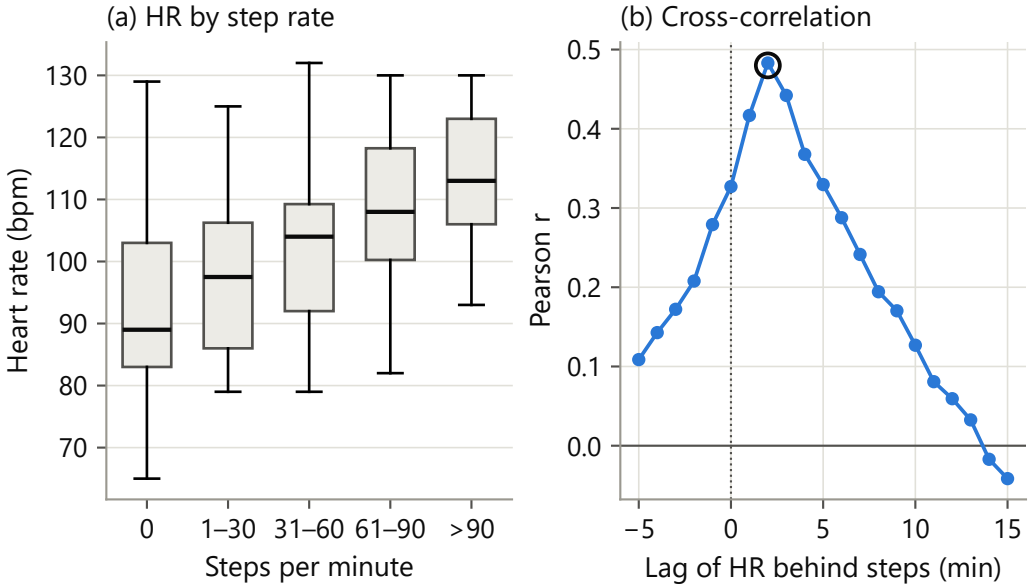


Figure 17: Heart-rate response to stepping (wrist band, 393 awake minutes). (a) Heart rate grouped by the number of steps in the same minute (minutes per group: 0: 320, 1–30: 28, 31–60: 22, 61–90: 10, >90: 13); Spearman $\rho = 0.28$. (b) Correlation between heart rate and step count shifted by -5 to $+15$ min; the strongest association ($r = 0.48$) occurs when heart rate lags steps by 2 min (circled). Minute samples are autocorrelated, so no p-values are reported.

7 Drinking

The bottle log has 1,075 mL: 500 mL at 00:06, 500 mL at 19:09 and 75 mL at 20:02 (Fig. 18). Between the drink just after midnight and 19:09 nothing was logged for 19 hours, a period that included the midday outing, the afternoon sitting and the rise of heart rate to 102–112 bpm in the late afternoon; all of the day's water was drunk just after midnight or in the evening. Her logged intake was 599 mL on day 2 and 1,075 mL on day 3, below the EFSA adequate intake for women (about 2 L/day including food, of which roughly 1.6 L from drinks); if some drinks were not logged, these are minimum values.

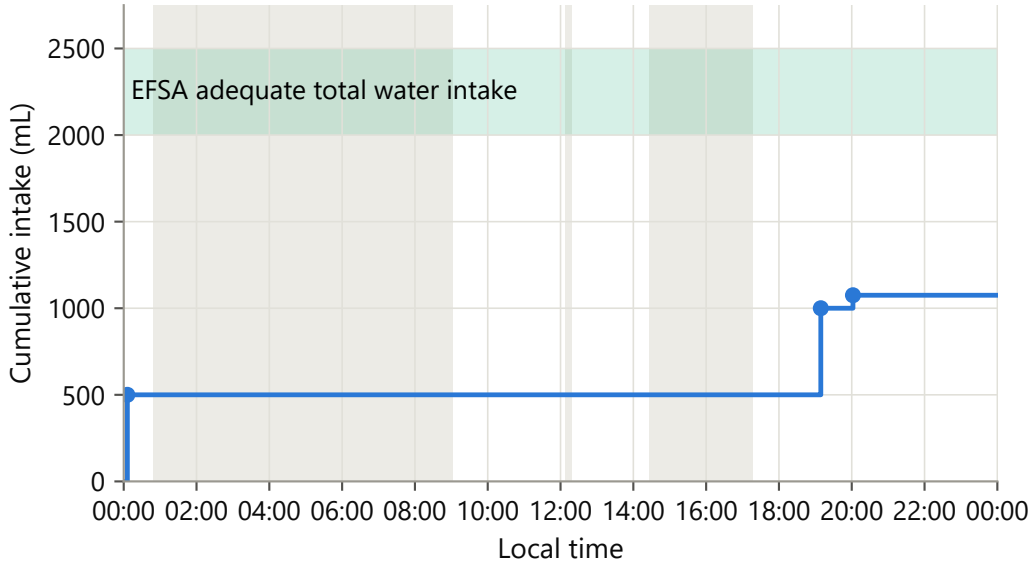


Figure 18: Cumulative fluid intake from the smart-bottle log (3 drinks, 1075 mL). Grey bands mark the sensor sessions; the shaded band is the EFSA adequate intake for total water (2.0–2.5 L/day, including water from food). 500 mL was consumed before the first walk. Longest gap without drinking: 19 h (00:06–19:09).

8 Days 1–3

Her gait was consistent over the three days: cadence 105–118 steps/min, strides of 1.0–1.1 s and a left-foot pressure stance of 62–65 %. All three rickshaw rides had $PM_{2.5}$ of 74–98 $\mu g/m^3$ and 75–80 dBA, more particles than the walks around them (Figs. 19 and 20). The clearest left-right pattern is

Table 4: Spriha's three days (measures that are valid for her).

Measure	Day 1	Day 2	Day 3
Walk cadence (steps/min)	116–118	105–108	107–114
Stride time (s)	1.02–1.04	1.10–1.14	1.06–1.13
Stance, left pressure sensors (%)	62	65	62
Foot-tilt range, left / right (°)	51 / 48	55 / 51	55–64 / 46–48
Awake heart rate, mean (bpm)	94	96	95
Heart rate in the rickshaw (bpm)	95	118–122	88–94
Rickshaw ride: $PM_{2.5}$ ($\mu g/m^3$) / sound (dBA)	98 / 80	74 / 77	76 / 75
Upright or straight sitting (%)	–	99	79
Fluid logged (mL)	–	599	1,075
Band sleep record (h)	–	3.2*	2.0*

*incomplete: the band recorded only part of the night.

the larger rotation of the left foot on all three days.

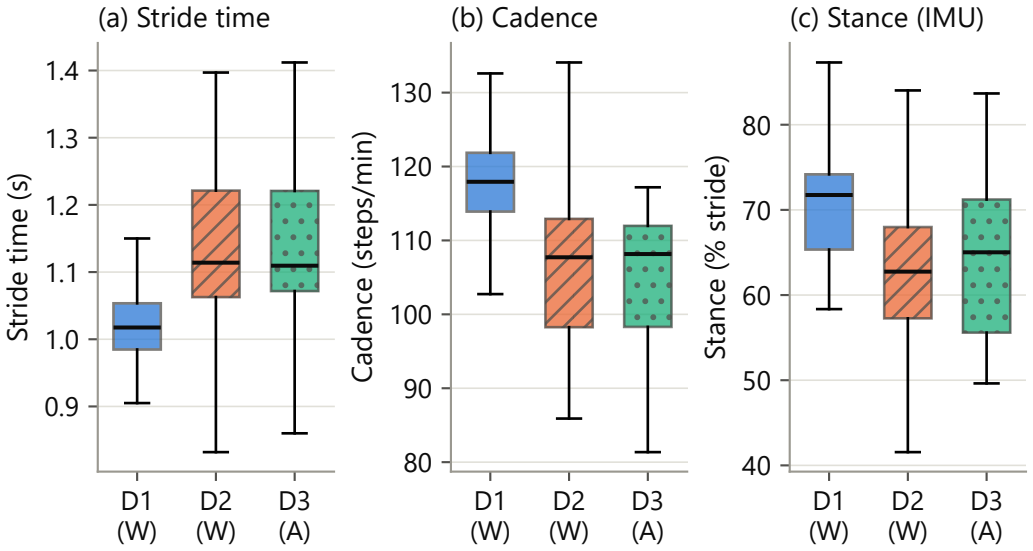


Figure 19: Right-foot gait during the main walk of each day (steady walking only: bouts of at least 10 strides without their first and last two strides; day 1: session W, 192 steady strides; day 2: session W, 164 steady strides; day 3: session A, 56 steady strides). Cadence is 120 divided by stride time. Stance is measured from IMU gait events (heel-strike impact to IMU toe-off), which stay valid after the right first-metatarsal pressure sensor was replaced (day 5 onwards).

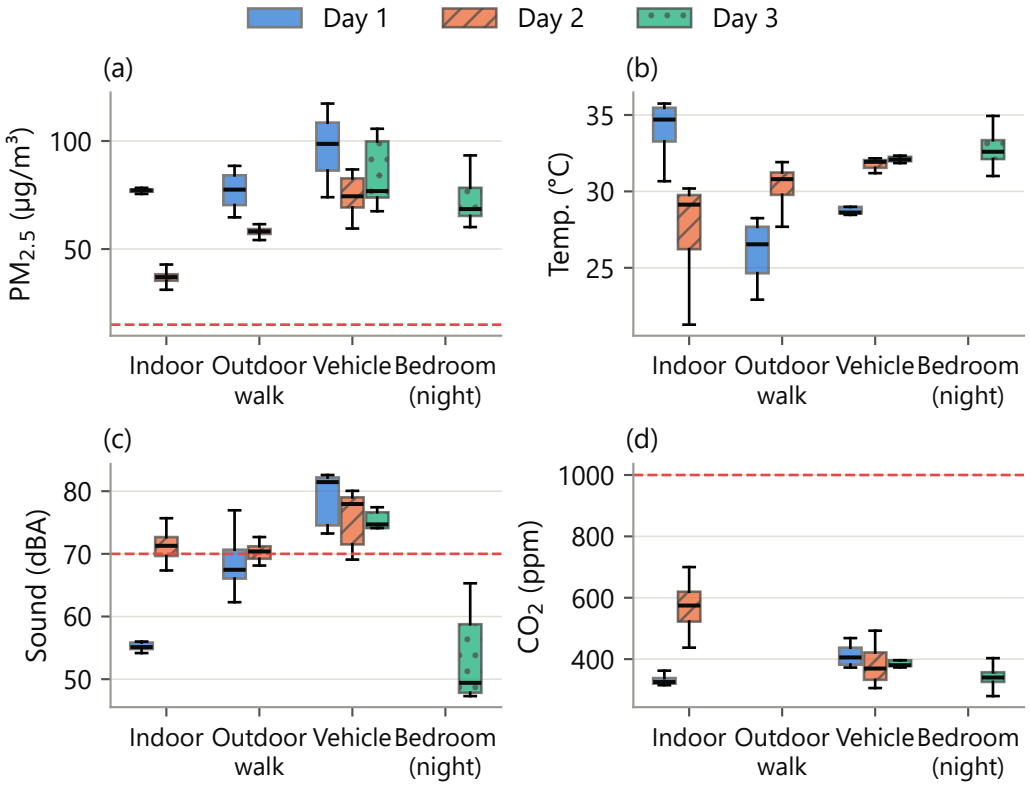


Figure 20: Environment-node readings (1-min means) by setting and day: (a) $PM_{2.5}$, (b) temperature, (c) sound level and (d) CO_2; outdoor CO_2 readings below 400 ppm are excluded as implausible. Dashed lines: WHO $PM_{2.5}$ 24-h guideline, 70 dBA, 1000 ppm. A setting is missing on days without that recording. Temperatures include a -5 °C self-heating correction.

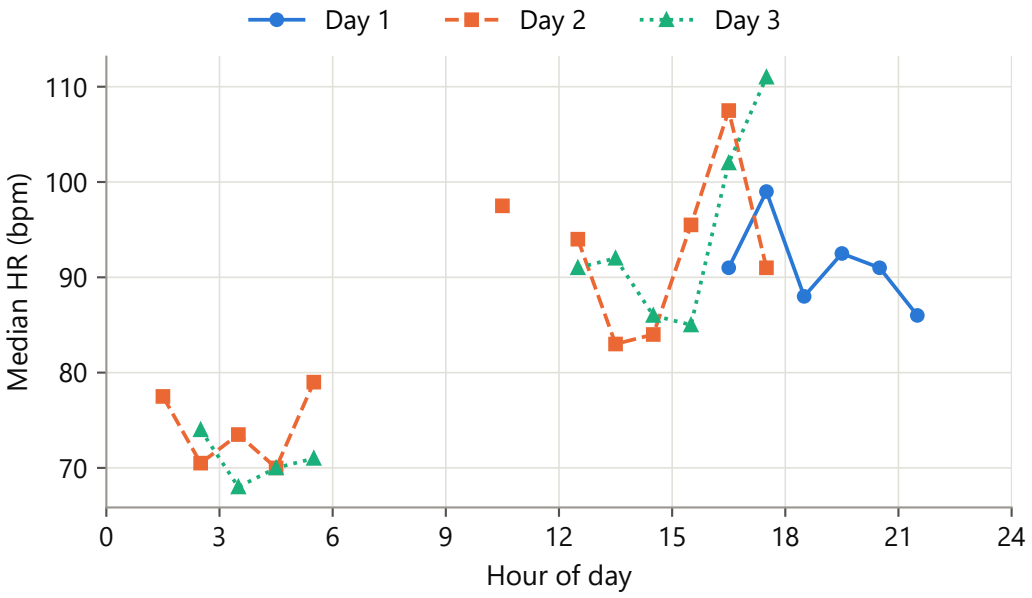


Figure 21: Hour-by-hour median heart rate (wrist band) on each day. Hours without band data are not drawn.

9 Comparison with participant 1

- **Gait** (Fig. 22): over her three days, shorter and quicker strides (median stride time 1.08 s, cadence 112 steps/min) than participant 1 (1.18 s, 102 steps/min); stance from the left pressure sensors, which are original for both, is the same (64 vs 64 %).
- **Settings** (Fig. 23): lower but louder indoor $PM_{2.5}$ (37 vs 52 $\mu g/m^3$; 71 vs 63 dBA), a hotter and quieter bedroom (33 vs 29 °C; 49 vs 62 dBA), similar outdoor $PM_{2.5}$ (60 vs 68).
- **Rickshaw rides**: her three rides had median $PM_{2.5}$ of 74–98 $\mu g/m^3$, against 53–56 on participant 1's two rickshaw rides; heart rate in vehicles was 113 bpm for her and 137 bpm for participant 1 (largely his metro evening).
- **Drinking**: her logged intake (0.6 and 1.1 L on two days) was at the low end of participant 1's range (0.9–2.2 L per day), and on both of her days most of it came at night or in the evening.

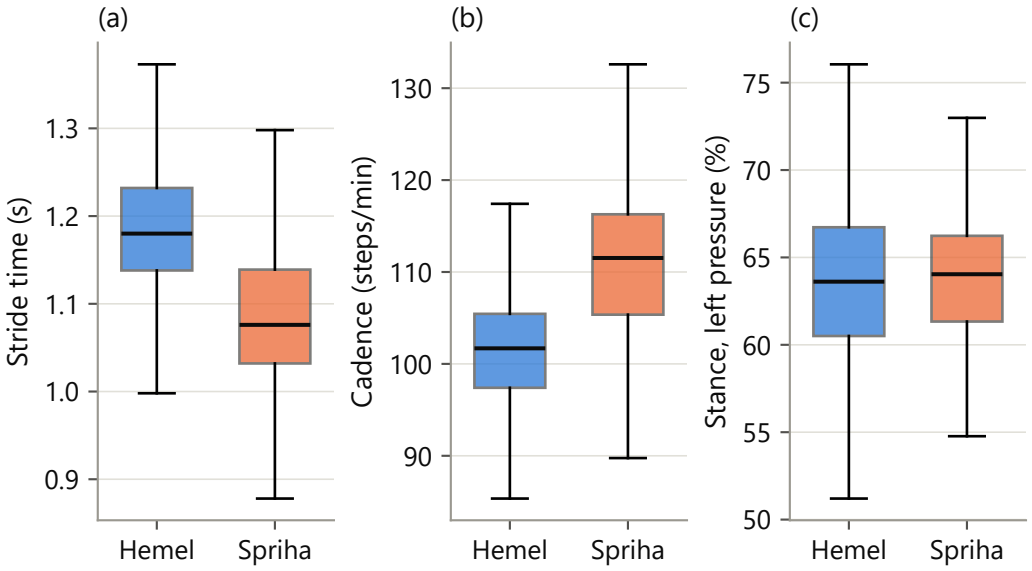


Figure 22: Gait of the two participants in all walks (steady strides; Hemel: days 1–8, 16,161 strides; Spriha: days 1–3, 927 strides). (a) Stride time, (b) cadence and (c) stance phase from the left insole's pressure sensors (the left insole's sensors were original and healthy for both participants).

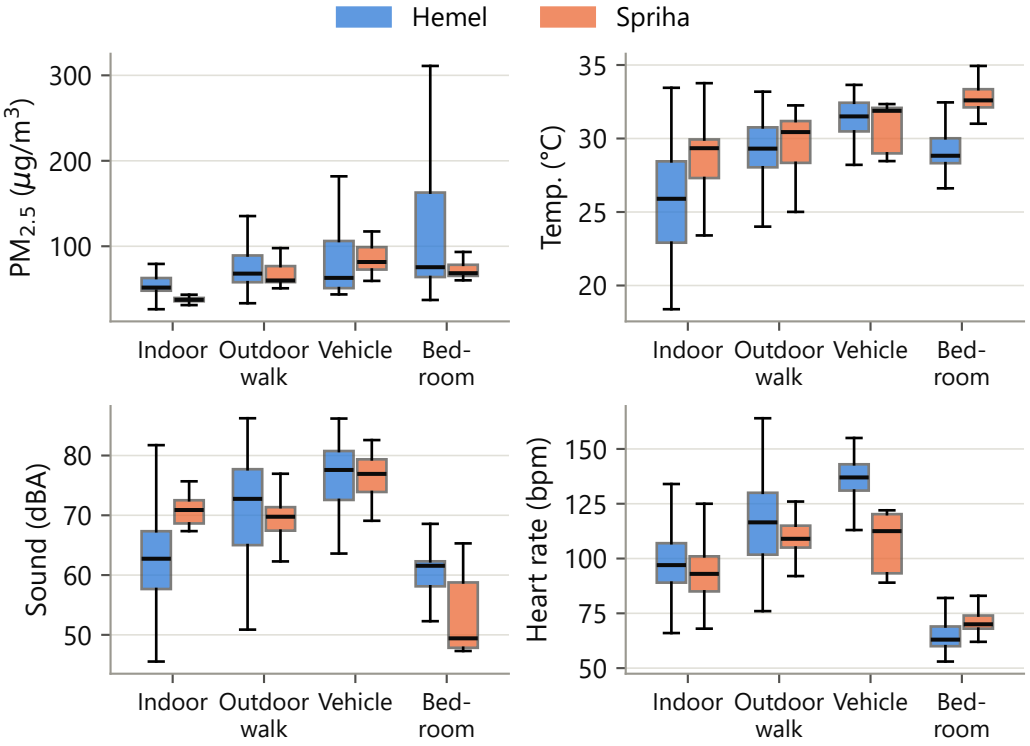


Figure 23: Environment and heart rate by setting for the two participants (1-minute values; Hemel days 1–8, Spriha days 1–2). Temperatures include the −5 °C correction. Spriha's band records heart rate every 10 minutes, so her heart-rate boxes rest on far fewer minutes.

10 Sensor-system notes

1. **Duplicate uploads**: the morning and afternoon folders contain the same recording; if there was a separate afternoon outing, it was not uploaded.

2. **Half-rate right insole** (62 samples/s, queued packets) on the third day running; inter-insole timing is not valid.

3. **Replaced right sensors**: the heel replacement did not unload during swing today; pressure-based measures of the right foot are not used.

4. **Band**: heart rate is now recorded every minute, which made the heart-rate analyses comparable with participant 1. Sleep is still recorded only in part; wearing the band snugly at night may help.

5. **Notes**: the rides were again recognised from the sensors; short notes of rides and drinks would confirm them.

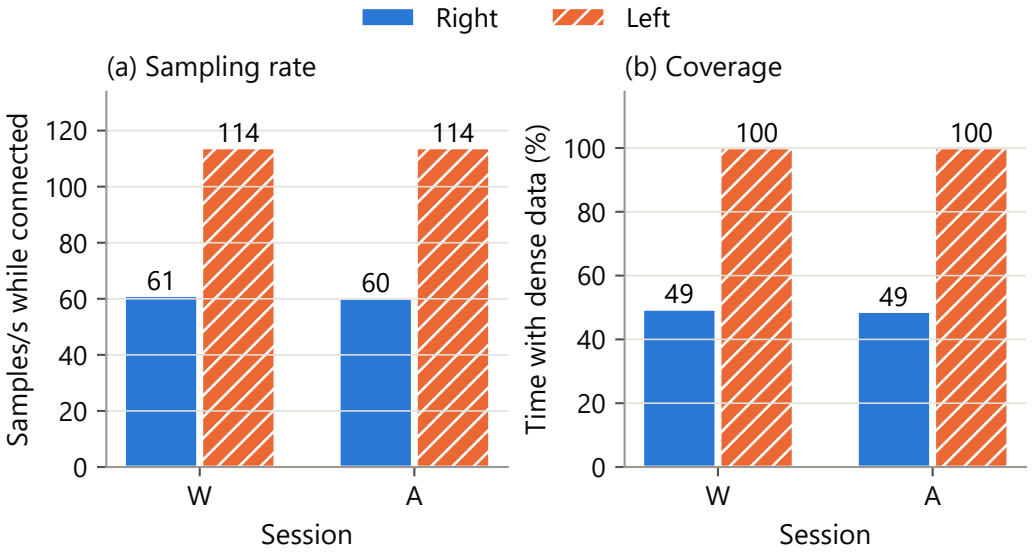


Figure 24: Data delivered by the right and left insoles in each session with both feet: (a) average sampling rate while the node was connected (nominal about 112 Hz) and (b) share of the recording covered by densely sampled data (consecutive samples $\leq$20 ms apart). The left insole's low coverage limits the left/right comparison to cycle-averaged timing and shape.